Med Term Quiz #2 — Immunity & Blood

Study guide for NRSG 231 Med Terminology Quiz #2 (due Oct 14, 2026). Focus: blood and immune/lymphatic terminology from the med-term textbook (cardiovascular/blood chapter; lymphatic/immune chapter) plus Pathophysiology terminology from Blood, Immune 1, and Immune 2 lectures. **Spelling counts.** Source tags: [handout] = on the course study-guide handout; [general] = standard medical terminology, not on that handout. Handout typos noted where relevant.

Textbook (Ehrlich Ch 5 blood / Ch 6)

From Ehrlich et al., *Medical Terminology for Health Professions* 8th ed. Ch 5 blood section and Ch 6 Lymphatic & Immune Systems. Page cites are printed book pages. Items in the later sections marked ✓ **Ehrlich** are confirmed by the textbook.

Ch 5 — Blood

Word parts (vocab list)

Term/Part	Meaning	Page
angi/o	blood or lymph vessel	[Ehrlich p. 123]
aort/o	aorta	[Ehrlich p. 123]
arteri/o	artery	[Ehrlich p. 123]
ather/o	plaque, fatty substance	[Ehrlich p. 123]
brady-	slow	[Ehrlich p. 123]
cardi/o	heart	[Ehrlich p. 123]
-crasia	a mixture or blending	[Ehrlich p. 123]
-emia	blood, blood condition	[Ehrlich p. 123]
erythr/o	red	[Ehrlich p. 123]
hem/o, hemat/o	blood, relating to the blood	[Ehrlich p. 123]
leuk/o	white	[Ehrlich p. 123]
phleb/o	vein	[Ehrlich p. 123]
tachy-	fast, rapid	[Ehrlich p. 123]
thromb/o	clot	[Ehrlich p. 123]
ven/o	vein	[Ehrlich p. 123]

Key terms

Term/Part	Meaning	Page
Blood	the fluid tissue in the body; composed of 55% liquid plasma and 45% formed elements; transports oxygen and nutrients to body tissues, returns waste products to the kidneys, carries carbon dioxide to the lungs	[Ehrlich p. 132]
Plasma	a straw-colored fluid that contains nutrients, hormones, and waste products; 91% water; the remaining 9% consists mainly of proteins, including the clotting proteins	[Ehrlich p. 133]
Serum	plasma fluid after the blood cells and the clotting proteins have been removed	[Ehrlich p. 133]
Fibrinogen	clotting protein found in plasma; has an important role in clot formation to control bleeding	[Ehrlich p. 133]

Term/Part	Meaning	Page
Prothrombin	clotting protein found in plasma; has an important role in clot formation to control bleeding	[Ehrlich p. 133]
Formed elements	the formed elements of blood include erythrocytes, leukocytes, and thrombocytes	[Ehrlich p. 133]
Erythrocytes (red blood cells, RBCs)	mature red blood cells produced by the red bone marrow (erythr/o means red, and -cytes means cells); the primary role of these cells is to transport oxygen to the tissues	[Ehrlich p. 133]
Hemoglobin	the oxygen-carrying blood protein pigment of the erythrocytes (hem/o means blood, and -globin means protein)	[Ehrlich p. 133]
Leukocytes (white blood cells, WBCs)	the blood cells involved in defending the body against infective organisms and foreign substances (leuk/o means white, and -cytes means cells)	[Ehrlich p. 133]
Neutrophils	formed in red bone marrow; the most common type of WBC; through phagocytosis, play a major role in the immune system's defense against pathogens, including bacteria, viruses, and fungi	[Ehrlich p. 133]
Basophils	formed in red bone marrow; the least common type of WBC; responsible for causing the symptoms of allergies	[Ehrlich p. 133]
Eosinophils	formed in red bone marrow and then migrate to tissues throughout the body; destroy parasitic organisms and play a major role in allergic reactions	[Ehrlich p. 133]
Phagocytosis	the process of destroying pathogens by surrounding and swallowing them	[Ehrlich p. 133]
Lymphocytes	formed in red bone marrow, in lymph nodes, and in the spleen; identify foreign substances and germs (bacteria or viruses) in the body and produce antibodies that specifically target them	[Ehrlich p. 134]
Monocytes	formed in red bone marrow, lymph nodes, and the spleen; through phagocytosis, provide immunological defenses against many infectious organisms	[Ehrlich p. 134]
Thrombocytes (platelets)	the smallest formed elements of the blood; play an important role in the clotting of blood (thromb/o means clot, and -cytes means cells); when a vessel is damaged they become sticky and clump together to form a clot to stop the bleeding	[Ehrlich p. 134]
Blood types	classified according to the presence or absence of certain antigens; the four major blood types are A, AB, B, and O; type is listed as type followed by Rh factor (such as A– or O+)	[Ehrlich p. 134]
Antigen	any substance that the body regards as being foreign	[Ehrlich p. 134]
Rh factor	defines the presence or absence of the Rh antigen on red blood cells; so named because this antigen was first found in rhesus monkeys; about 85% of Americans are Rh positive (Rh+) and 15% are Rh negative (Rh–)	[Ehrlich p. 134]
Blood gases	gases that are normally dissolved in the liquid portion of blood; the major blood gases are oxygen (O2), carbon dioxide (CO2), and nitrogen (N2)	[Ehrlich p. 135]

Term/Part	Meaning	Page
Hematologist	a physician who specializes in diagnosing and treating abnormalities, diseases, and disorders of the blood and blood-forming tissues (hemat means blood, and -ologist means specialist)	[Ehrlich p. 135]
Hemolysis	the breaking down of red blood cells (hemolytic means pertaining to hemolysis)	[Ehrlich p. 144]
Hemostasis	listed as a primary term in the chapter vocabulary list; its definition does not appear in the pages provided for this study guide	[Ehrlich p. 123]

Pathology

Term/Part	Meaning	Page
Blood dyscrasia	any pathologic condition of the cellular elements of the blood (dys- means bad, and -crasia means a mixture or blending)	[Ehrlich p. 142]
Hemochromatosis (iron overload disease)	a genetic disorder in which the intestines absorb too much iron (hem/o means blood, chromat means color, and -osis means abnormal condition or disease); the excess iron enters the bloodstream and accumulates in organs, where it causes damage	[Ehrlich p. 142]
Leukopenia	a decrease in the number of disease-fighting white blood cells circulating in the blood (leuk/o means white, and -penia means deficiency); also known as a low white blood cell count	[Ehrlich p. 142]
Polycythemia	an abnormal increase in the number of red cells in the blood due to excess production of these cells by the bone marrow	[Ehrlich p. 143]
Sepsis (septicemia)	a potentially life-threatening infection that results from bacteria or other infectious organisms entering the bloodstream; sometimes referred to as blood poisoning	[Ehrlich p. 143]
Thrombocytopenia	a condition in which there is an abnormally small number of platelets circulating in the blood (thromb/o means clot, cyt/o means cell, and -penia means deficiency); sometimes associated with abnormal bleeding	[Ehrlich p. 143]
Thrombocytosis	an abnormal increase in the number of platelets in the circulating blood (thromb/o means clot, cyt means cell, and -osis means abnormal condition)	[Ehrlich p. 143]
Hemorrhage	the loss of a large amount of blood in a short time (hem/o means blood, and -rrhage means bleeding)	[Ehrlich p. 143]

Term/Part	Meaning	Page
Transfusion reaction	a serious and potentially fatal complication of a blood transfusion in which a severe immune response occurs because the patient's blood and the donated blood do not match	[Ehrlich p. 143]
Cholesterol	a fatty substance that travels through the blood and is found in all parts of the body; aids in the production of cell membranes, some hormones, and vitamin D; excessively high levels of certain types of cholesterol can lead to heart disease	[Ehrlich p. 143]
Hyperlipidemia	the general term used to describe elevated levels of cholesterol and other fatty substances in the blood (hyper- means excessive, lipid means fat, and -emia means blood condition)	[Ehrlich p. 143]
Myelodysplastic syndrome	a type of cancer in which there is insufficient production of one or more types of blood cells due to dysfunction of the bone marrow; some patients later develop acute myeloid leukemia	[Ehrlich p. 144]
Leukemia	a type of cancer characterized by a progressive increase in the number of abnormal leukocytes (white blood cells) found in blood-forming tissues, other organs, and in the circulating blood (leuk means white, and -emia means blood condition); the four most common types are acute or chronic myeloid leukemia and acute or chronic lymphoblastic leukemia	[Ehrlich p. 144]
Anemia	a lower-than-normal number of erythrocytes (red blood cells) in the blood (an- means without or less than, and -emia means blood condition) causing fatigue and shortness of breath; severity is usually measured by a decrease in the amount of hemoglobin in the blood	[Ehrlich p. 144]
Aplastic anemia	characterized by an absence of all formed blood elements caused by the failure of blood cell production in the bone marrow (a- means without, plast means growth, and -ic means pertaining to); anemia (low RBCs) leads to fatigue and weakness, leukopenia (low WBCs) causes increased risk of infection, thrombocytopenia (low platelets) results in bleeding	[Ehrlich p. 144]
Iron-deficiency anemia	the most common form of anemia; without sufficient iron to help create hemoglobin, blood cannot carry oxygen effectively; can be caused by blood loss, poor absorption of iron, or inadequate dietary intake of iron	[Ehrlich p. 144]
Megaloblastic anemia	a blood disorder characterized by anemia in which the red blood cells are larger than normal (megalo- means large, blast means immature, and -ic means pertaining to); usually results from a deficiency of folic acid or vitamin B12	[Ehrlich p. 144]
Pernicious anemia	caused by a lack of the protein intrinsic factor (IF) that helps the body absorb vitamin B12 from the gastrointestinal tract; vitamin B12 is necessary for the formation of red blood cells	[Ehrlich p. 144]
Hemolytic anemia	characterized by an inadequate number of circulating red blood cells due to the premature destruction of red blood cells by the spleen (hem/o means relating to blood, and -lytic means to destroy)	[Ehrlich p. 144]

Term/Part	Meaning	Page
Sickle cell anemia	a serious genetic disorder that causes abnormal hemoglobin, resulting in some red blood cells assuming an abnormal sickle shape; this shape interferes with normal blood flow, resulting in damage to most of the body systems	[Ehrlich p. 144]
Thalassemia	an inherited blood disorder that causes mild or severe anemia due to reduced hemoglobin and fewer red blood cells than normal	[Ehrlich p. 144]

Diagnostic procedures

Term/Part	Meaning	Page
Blood tests	“Blood tests, computed tomography (CT) scans, MRIs, and ultrasonic diagnostic procedures are discussed in Chapter 15” (not detailed in this chapter)	[Ehrlich p. 145]
Cross-matching blood for transfusions	the Rh factor is an important consideration in cross-matching blood for transfusions (see Chapter 15)	[Ehrlich p. 135]
Cholesterol blood test — total cholesterol (Table 5.2)	total cholesterol is measured in terms of milligrams (mg) per deciliter (dL) of blood; desirable levels are below 200 mg/dL, borderline high 200–239 mg/dL, high 240 mg/dL and above	[Ehrlich p. 143]
Low-density lipoprotein cholesterol (LDL)	referred to as bad cholesterol because excess quantities of LDL contribute to plaque buildup in the arteries; optimal levels are below 100 mg/dL	[Ehrlich p. 143]
High-density lipoprotein cholesterol (HDL)	referred to as good cholesterol because it carries unneeded cholesterol back to the liver for processing and does not contribute to plaque buildup; low levels are below 40 mg/dL, high levels are 60 mg/dL and above	[Ehrlich p. 143]
Triglycerides	combinations of fatty acids attached to glycerol that are also found normally in the blood in limited quantities; normal levels are below 150 mg/dL	[Ehrlich p. 143]

Treatments

Term/Part	Meaning	Page
Anticoagulant	slows coagulation and prevents new clots from forming (coagulation is the process of clotting blood); although sometimes called blood thinners, they do not actually make the blood thinner	[Ehrlich p. 147]
Coumadin® (warfarin)	an anticoagulant administered to prevent blood clots from forming or growing larger; often prescribed for patients with clotting difficulties, certain types of heartbeat irregularities, or after a heart attack or heart valve replacement surgery	[Ehrlich p. 147]
Heparin	another common anticoagulant; works faster than Coumadin, but must be administered by injection or in an IV solution	[Ehrlich p. 147]
Aspirin (baby aspirin, 81 mg)	a very small daily dose may be recommended to reduce the risk of a heart attack or stroke by reducing the ability of the blood to clot	[Ehrlich p. 147]

Term/Part	Meaning	Page
Cholesterol-lowering drugs	used to combat hyperlipidemia by reducing the undesirable cholesterol levels in the blood	[Ehrlich p. 147]
Thrombolytic (clot-busting drug)	dissolves or causes a thrombus to break up (thromb/o means clot, and -lytic means to destroy)	[Ehrlich p. 147]
Thrombolytic therapy	the use of drugs to break up a thrombus	[Ehrlich p. 147]
Tissue plasminogen activator (tPA)	a thrombolytic administered to some patients having a heart attack or stroke; if administered within a few hours after symptoms begin, this medication can dissolve the damaging blood clots	[Ehrlich p. 147]
Blood transfusion	mentioned as a treatment; the Rh factor is an important consideration in cross-matching blood for transfusions (see Chapter 15)	[Ehrlich p. 135]
Nutritional interventions (cholesterol-limited diet)	patients with high cholesterol or other cardiac disease might be prescribed a diet limited in cholesterol; all foods that come from animals (dairy, poultry, seafood, meat, eggs) contain cholesterol	[Ehrlich p. 148]

Textbook vs course handout

- **An-** — handout: “without, not”; textbook (anemia): “an- means without or less than.” The textbook wording adds “less than” and does not include “not.”
- **Myel-** — handout: “spinal cord”; in this chapter myelodysplastic syndrome is “insufficient production of one or more types of blood cells due to dysfunction of the bone marrow,” so the textbook uses myel- here for bone marrow, which the handout does not list.
- **-cyte** — handout: “-cyte = cells in”; textbook writes the plural “-cytes means cells” (erythrocytes, leukocytes, thrombocytes). Meaning wording differs (“cells in” vs. “cells”).
- **-oma** — handout: “tumor of”; textbook (atheroma): “-oma means tumor.”
- **-rrhage** — handout: “bleeding, abnormal excessive fluid discharge”; textbook (hemorrhage): “-rrhage means bleeding.”
- **Word parts the handout omits** (used in this chapter’s blood terms): **-emia** “blood, blood condition”; **hem/o, hemat/o** “blood, relating to the blood”; **erythr/o** “red”; **thromb/o** “clot”; **-crasia** “a mixture or blending.”

Ch 6 — Lymphatic & Immune Systems

Word parts (vocab list)

Term/Part	Meaning (textbook wording, concise)	Page
anti-	Against.	[Ehrlich p. 164]
carcin/o	Cancerous.	[Ehrlich p. 164]
immun/o	Immune, protection, safe.	[Ehrlich p. 164]
lymph/o	Lymph, lymphatic tissue.	[Ehrlich p. 164]
lymphaden/o	Lymph node or gland.	[Ehrlich p. 164]
lymphangi/o	Lymph vessel.	[Ehrlich p. 164]
neo-, ne/o	New, strange.	[Ehrlich p. 164]
-oma	Tumor, neoplasm.	[Ehrlich p. 164]
onc/o	Tumor.	[Ehrlich p. 164]
phag/o	Eat, swallow.	[Ehrlich p. 164]
-plasm	Formative material of cells.	[Ehrlich p. 164]
sarc/o	Flesh, connective tissue.	[Ehrlich p. 164]

Term/Part	Meaning (textbook wording, concise)	Page
splen/o	Spleen.	[Ehrlich p. 164]
-tic	Pertaining to.	[Ehrlich p. 164]
tox/o	Poison, poisonous.	[Ehrlich p. 164]

Key terms

Term/Part	Meaning (textbook wording, concise)	Page
lacteals	Specialized lymphatic structures that absorb fats that cannot be transported by the bloodstream.	[Ehrlich p. 165]
interstitial fluid	Also intercellular or tissue fluid; plasma from arterial blood that flows into spaces between tissue cells.	[Ehrlich p. 165]
lymph	Clear, watery fluid containing electrolytes and proteins; the remaining 10% of returning interstitial fluid. Removes dead cells, debris, and pathogens.	[Ehrlich p. 165]
lymphatic capillaries	Microscopic, blind-ended tubes near the body surface, with walls one cell thick; admit lymph.	[Ehrlich p. 166]
lymphatic vessels	Progressively larger vessels deeper in tissues; valves prevent backward lymph flow.	[Ehrlich p. 166]
right lymphatic duct	Collects lymph from the right head and neck, upper right body quadrant, and right arm; empties into the right subclavian vein.	[Ehrlich p. 166]
thoracic duct	Largest lymphatic vessel; collects lymph from the left head and neck, upper left trunk, left arm, lower trunk, and both legs; empties into the left subclavian vein.	[Ehrlich p. 166]
lymph nodes	Small, bean-shaped structures containing lymphocytes that destroy pathogens; filter impurities from lymph.	[Ehrlich p. 166]
cervical lymph nodes	Located along the sides of the neck.	[Ehrlich p. 166]
axillary lymph nodes	Located under the arms in the armpits.	[Ehrlich p. 166]
inguinal lymph nodes	Located in the groin area of the lower abdomen.	[Ehrlich p. 166]
bone marrow	Produces lymphocytes, specialized leukocytes.	[Ehrlich p. 163]
lymphocytes	Also lymphoid cells; leukocytes formed in bone marrow as stem cells that defend the body against antigens.	[Ehrlich p. 166]
antigen	Any substance the body regards as foreign, including viruses, bacteria, toxins, and transplanted tissues.	[Ehrlich p. 168]
natural killer cells	Kill cancer cells and cells infected by viruses.	[Ehrlich p. 168]
B cells	Also B lymphocytes; specialized lymphocytes that produce antibodies.	[Ehrlich p. 168]
plasma cells	Develop from B cells and secrete large volumes of antibodies coded to destroy specific antigens.	[Ehrlich p. 168]
T cells	Also T lymphocytes; originate in the thymus and play a central role in cell-mediated immunity.	[Ehrlich p. 168]
cytokines	Proteins such as interferons and interleukins released primarily by T cells; act as intracellular signals to begin the immune response.	[Ehrlich p. 168]

Term/Part	Meaning (textbook wording, concise)	Page
interferons	Produced in response to antigens, particularly viruses or tumor cells; activate immunity, slow or stop viral multiplication, and signal other cells to increase defenses.	[Ehrlich p. 168]
interleukins	Direct B and T cells to divide and proliferate, among other immune roles.	[Ehrlich p. 168]
lymphoid	Pertaining to the lymphatic system or resembling lymph or lymphatic tissue.	[Ehrlich p. 168]
tonsils	Three masses of lymphoid tissue forming a protective ring around the back of the nose and upper throat.	[Ehrlich p. 168]
adenoids	Also nasopharyngeal tonsils; located in the nasopharynx.	[Ehrlich p. 168]
palatine tonsils	Located on the left and right sides of the throat, visible at the back of the mouth.	[Ehrlich p. 168]
lingual tonsils	Located at the base of the tongue; not readily visible.	[Ehrlich p. 169]
thymus	Lymphoid tissue above the heart; secretes a hormone stimulating lymphocyte maturation into T cells.	[Ehrlich p. 169]
vermiform appendix	Hangs from the lower cecum; may play an important immune-system role.	[Ehrlich p. 169]
spleen	Sac-like lymphoid mass in the left upper abdomen; filters blood, forms lymphocytes and monocytes, and destroys worn-out erythrocytes.	[Ehrlich p. 169]
hemolytic	Describes the spleen’s destruction of worn-out erythrocytes and release of their hemoglobin for reuse.	[Ehrlich p. 169]
toxins	Poisonous or harmful substances.	[Ehrlich p. 170]
antigen–antibody reaction	Also immune reaction; binding antigens to antibodies, labeling dangerous antigens for recognition and destruction.	[Ehrlich p. 171]
tolerance	Acquired unresponsiveness to a specific antigen; also a decline in effective drug response, usually from repeated use.	[Ehrlich p. 171]
antibody	Disease-fighting protein created by the immune system in response to a specific antigen.	[Ehrlich p. 171]
immunoglobulins	Also antibodies; secreted by plasma cells and bind specific antigens.	[Ehrlich p. 171]
Immunoglobulin G (IgG)	Most abundant antibody class; in serum and lymph; active against bacteria, fungi, viruses, and foreign particles.	[Ehrlich p. 171]
Immunoglobulin A (IgA)	Produced predominantly against ingested antigens; in secretions; prevents viral and bacterial attachment to epithelial surfaces.	[Ehrlich p. 171]
Immunoglobulin M (IgM)	In circulating body fluids; first antibodies appearing after initial antigen exposure.	[Ehrlich p. 171]
Immunoglobulin D (IgD)	Found only on B-cell surfaces; important in B-cell activation.	[Ehrlich p. 171]
Immunoglobulin E (IgE)	Produced in lungs, skin, and mucous membranes; responsible for allergic reactions.	[Ehrlich p. 171]
phagocytes	Specialized leukocytes that destroy debris, dust, pollen, and pathogens by phagocytosis.	[Ehrlich p. 172]

Term/Part	Meaning (textbook wording, concise)	Page
phagocytosis	Destroying pathogens by surrounding and swallowing them.	[Ehrlich p. 172]
monocytes	Leukocytes providing immunological defenses; replenish macrophages and dendritic cells.	[Ehrlich p. 172]
macrophage	Leukocyte that surrounds and kills invading cells; removes dead cells and stimulates other immune cells.	[Ehrlich p. 172]
dendritic cells	Leukocytes that search for infection-producing antigens, swallow them, and alert B and T cells.	[Ehrlich p. 172]
complement system	Proteins normally circulating in inactive form; complement antibodies in dissolving and removing pathogenic bacteria and other foreign cells.	[Ehrlich p. 172]
immunity	State of being resistant to a specific disease.	[Ehrlich p. 172]
natural immunity	Also passive immunity; resistance present without antigen administration or disease exposure; present at birth and augmented by breast milk.	[Ehrlich p. 172]
acquired immunity	Obtained by having had a contagious disease; vaccination provides protection without the risk of having the disease.	[Ehrlich p. 172]
allergist	Specialist in diagnosing and treating altered immunologic reactivity, such as allergic reactions.	[Ehrlich p. 172]
immunologist	Specialist in diagnosing and treating immune-system disorders.	[Ehrlich p. 172]
lymphologist	Physician specializing in diagnosing and treating lymphatic-system disorders.	[Ehrlich p. 172]
oncologist	Physician specializing in diagnosing and treating malignant disorders.	[Ehrlich p. 172]
allergen	Substance producing an allergic reaction in an individual.	[Ehrlich p. 174]
histamine	Body-produced substance causing itching, sneezing, runny nose, and watery eyes in an allergic reaction.	[Ehrlich p. 174]
pathogen	Microorganism causing disease in humans.	[Ehrlich p. 177]
microorganism	Living organism visible only with a microscope.	[Ehrlich p. 177]
pathogenic	Capable of producing disease.	[Ehrlich p. 177]
bacteria	One-celled microscopic organisms; most are not harmful to humans.	[Ehrlich p. 177]
bacilli	Rod-shaped, spore-forming bacteria.	[Ehrlich p. 177]
rickettsia	Small bacterium living in lice, fleas, ticks, and mites.	[Ehrlich p. 177]
spirochetes	Long, slender, spiral-shaped bacteria with flexible walls; capable of movement.	[Ehrlich p. 177]
staphylococci	Bacteria forming irregular groups or clusters resembling grapes.	[Ehrlich p. 178]
Staphylococcus aureus	Also staph aureus; often infects wounds and causes serious problems such as toxic shock syndrome or food poisoning.	[Ehrlich p. 178]

Term/Part	Meaning (textbook wording, concise)	Page
streptococci	Bacteria forming a chain; some cause serious illnesses.	[Ehrlich p. 178]
nosocomial	Hospital or clinic acquired.	[Ehrlich p. 178]
fungus	Simple parasitic organism; some are harmless and others pathogenic.	[Ehrlich p. 179]
yeast	A type of fungus.	[Ehrlich p. 179]
parasite	Plant or animal living on or within another living organism at that organism's expense.	[Ehrlich p. 179]
viruses	Very small infectious agents that live only by invading other cells.	[Ehrlich p. 179]
herpesviruses	Group including varicella zoster, Epstein-Barr, cytomegalovirus, and herpes simplex; causes various human diseases.	[Ehrlich p. 180]
oncology	Study of the prevention, causes, and treatment of tumors and cancer.	[Ehrlich p. 181]

Pathology

Term/Part	Meaning (textbook wording, concise)	Page
lymphadenitis	Inflammation of lymph nodes; commonly swollen glands.	[Ehrlich p. 172]
lymphadenopathy	Any disease process affecting a lymph node or nodes.	[Ehrlich p. 173]
lymphangioma	Benign tumor formed by an abnormal collection of lymphatic vessels due to congenital lymphatic malformation.	[Ehrlich p. 173]
ruptured spleen	Medical emergency in which the spleen's covering is torn, usually following a blow to the abdomen.	[Ehrlich p. 173]
splenorrhagia	Bleeding from the spleen.	[Ehrlich p. 173]
splenomegaly	Abnormal enlargement of the spleen.	[Ehrlich p. 173]
lymphedema	Tissue swelling from abnormal accumulation of lymph fluid; caused by lymphatic damage preventing proper drainage.	[Ehrlich p. 173]
primary lymphedema	Hereditary lymphatic condition; swelling begins in the feet and progresses upward along the legs.	[Ehrlich p. 173]
secondary lymphedema	Caused by lymphatic-vessel damage, most frequently from cancer treatment, surgery, trauma, or burns.	[Ehrlich p. 173]
allergic reaction	Immune response to a harmless allergen as though it were a dangerous invader.	[Ehrlich p. 173]
allergy	Also hypersensitivity; overreaction by the body to a particular antigen.	[Ehrlich p. 174]
localized allergic response	Also cellular response; redness, itching, and burning where skin contacts an allergen.	[Ehrlich p. 174]
systemic reaction	Severe, rapidly developing response to an allergen; also described as anaphylaxis or anaphylactic shock.	[Ehrlich p. 174]

Term/Part	Meaning (textbook wording, concise)	Page
anaphylaxis	Severe systemic allergic response; without prompt medical aid, death can occur within minutes.	[Ehrlich p. 174]
autoimmune disorder	Immune system produces antibodies against its own tissues, mistaking healthy cells, tissues, or organs for antigens.	[Ehrlich p. 175]
Rheumatoid arthritis	Autoimmune disorder affecting joints and connective tissue.	[Ehrlich p. 175]
Myasthenia gravis	Autoimmune disorder affecting nerve and muscle synapses.	[Ehrlich p. 175]
Pernicious anemia	Autoimmune disorder affecting red blood cells.	[Ehrlich p. 175]
Crohn's disease	Autoimmune disorder affecting the intestines, ileum, or colon.	[Ehrlich p. 175]
Multiple sclerosis	Autoimmune disorder affecting the brain and spinal cord.	[Ehrlich p. 175]
Scleroderma	Autoimmune disorder affecting skin and connective tissues.	[Ehrlich p. 175]
Graves' disease	Autoimmune disorder affecting the thyroid gland.	[Ehrlich p. 175]
immunodeficiency disorder	Condition in which the immune response is compromised: weakened or not functioning properly.	[Ehrlich p. 175]
severe combined immunodeficiency	Inherited immune abnormalities causing increased susceptibility to infection and failure to thrive from infections.	[Ehrlich p. 175]
human immunodeficiency virus	Bloodborne infection in which the virus damages or kills T cells, progressively impairing immunity.	[Ehrlich p. 176]
opportunistic infection	Infection caused by a pathogen that does not normally cause illness in healthy humans but can infect a debilitated host.	[Ehrlich p. 176]
acquired immunodeficiency syndrome	Most advanced and fatal stage of an HIV infection.	[Ehrlich p. 176]
Kaposi's sarcoma	Cancer frequently associated with HIV; causes abnormal tissue patches under skin, in mucosal linings, or in other organs.	[Ehrlich p. 176]
anthrax	Contagious disease transmitted through livestock infected with Bacillus anthracis.	[Ehrlich p. 177]
Rocky Mountain spotted fever	Caused by a rickettsia transmitted by an infected tick's bite.	[Ehrlich p. 177]
Lyme disease	Tick-transmitted infection involving Borrelia burgdorferi; fever, headache, fatigue, and erythema migrans rash; may spread to joints, heart, and nervous system.	[Ehrlich p. 178]
syphilis	Disease caused by spirochetes; discussed in Chapter 14.	[Ehrlich p. 178]
antibiotic-resistant bacteria	Bacteria surviving antibiotic treatment become resistant to that particular drug.	[Ehrlich p. 178]
Methicillin-resistant Staphylococcus aureus	MRSA; bacteria resistant to most antibiotics; initial small red bumps with black tops become abscesses.	[Ehrlich p. 178]

Term/Part	Meaning (textbook wording, concise)	Page
Tinea pedis	Athlete’s foot; fungal infection developing between the toes.	[Ehrlich p. 179]
candidiasis	Yeast infection of skin or mucous membranes in warm, moist areas, caused by Candida albicans.	[Ehrlich p. 179]
oral thrush	Yeast infection occurring in the mouth.	[Ehrlich p. 179]
vaginal candidiasis	Yeast infection occurring in the vagina.	[Ehrlich p. 179]
malaria	Caused by a parasite transferred through an infected mosquito’s bite; fever, shaking, chills, headache, muscle aches, and fatigue.	[Ehrlich p. 179]
toxoplasmosis	Described as a parasite most commonly transmitted from pets through contaminated feces; can cause disease in a developing child.	[Ehrlich p. 179]
microcephalus	Abnormally small head and underdeveloped brain.	[Ehrlich p. 179]
hydrocephalus	Excess cerebrospinal fluid accumulates in brain ventricles.	[Ehrlich p. 179]
Ebola	Rare, generally fatal disease transmitted through infected bodily fluids when symptoms are present; late symptoms include massive internal bleeding and organ failure.	[Ehrlich p. 179]
influenza	Flu; highly contagious viral respiratory infection, usually occurring in seasonal epidemics.	[Ehrlich p. 179]
measles	Acute, highly contagious rubeola infection transmitted by respiratory droplets; red itchy rash, high fever, runny nose, and cough.	[Ehrlich p. 179]
mumps	Acute viral infection characterized by swollen parotid glands; can cause painful ovarian or testicular swelling in adults.	[Ehrlich p. 179]
rubella	German measles or three-day measles; viral infection with low fever, swollen glands, inflamed eyes, and fine pink rash; can cause fetal defects.	[Ehrlich p. 179]
rabies	Acute viral infection transmitted through an infected animal’s bite or saliva; almost always fatal once symptoms develop.	[Ehrlich p. 179]
West Nile virus	Spread by infected mosquitoes; mild disease has flu-like symptoms; severe disease spreads to spinal cord and brain.	[Ehrlich p. 179]
dengue fever	Mosquito-borne tropical disease caused by viruses in the Flavivirus genus.	[Ehrlich p. 180]
cytomegalovirus	CMV; found in most body fluids, often a silent infection; can cause serious illness with weakened immunity or congenital disabilities after fetal transmission.	[Ehrlich p. 180]
varicella	Chickenpox; highly contagious Varicella zoster infection with fever and itchy, fluid-filled blisters that burst and crust.	[Ehrlich p. 180]
herpes zoster	Shingles; reactivated dormant varicella virus causing painful skin eruptions along an inflamed nerve.	[Ehrlich p. 180]
postherpetic neuralgia	Complication following herpes zoster when nerve fibers are damaged; pain may be severe and persistent.	[Ehrlich p. 180]

Term/Part	Meaning (textbook wording, concise)	Page
infectious mononucleosis	Mono; caused by Epstein-Barr virus; fever, sore throat, enlarged lymph nodes, and possible spleen swelling or liver involvement.	[Ehrlich p. 180]; [Ehrlich p. 181]
tumor; neoplasm	Abnormal body-tissue growth with uncontrolled, abnormal, rapid, progressive cell multiplication.	[Ehrlich p. 181]
benign	Not life-threatening; a benign tumor is not cancer but can damage adjacent structures through pressure.	[Ehrlich p. 181]
myoma	Benign tumor made up of muscle tissue.	[Ehrlich p. 181]
malignant	Becoming progressively worse and life-threatening; a malignant tumor is cancer capable of distant spread.	[Ehrlich p. 181]
myosarcoma	Malignant tumor derived from muscle tissue.	[Ehrlich p. 181]
angiogenesis	Process through which a tumor creates its own blood supply to support growth.	[Ehrlich p. 181]
cancer	Diseases characterized by uncontrolled cell division and the ability to invade other tissues or spread to distant sites.	[Ehrlich p. 181]
metastasize	Spread from one place to another.	[Ehrlich p. 181]
metastasis	Process by which cancer spreads to a new site; also the resulting tumor.	[Ehrlich p. 181]
carcinoma	Malignant tumor occurring in epithelial tissue.	[Ehrlich p. 182]
carcinoma in situ	Malignant tumor in its original position that has not disturbed or invaded surrounding tissues.	[Ehrlich p. 182]
adenocarcinoma	Carcinoma derived from glandular tissue.	[Ehrlich p. 182]
sarcoma	Malignant tumor arising from hard, soft, or liquid connective tissues.	[Ehrlich p. 182]
hard-tissue sarcomas	Arise from bone or cartilage.	[Ehrlich p. 182]
osteosarcoma	Hard-tissue sarcoma usually involving the upper shaft of long bones, pelvis, or knee.	[Ehrlich p. 182]
soft-tissue sarcomas	Cancers of muscle, fat, fibrous tissue, blood and lymphatic vessels, or other supporting tissue.	[Ehrlich p. 182]
synovial sarcoma	Tumor of tissues surrounding a synovial joint.	[Ehrlich p. 182]
liquid-tissue sarcomas	Arise from blood and lymph.	[Ehrlich p. 182]
leukemia	Cancer of white blood-forming cells in bone marrow.	[Ehrlich p. 182]
basal cell carcinoma	Malignant tumor occurring in the epidermis’s basal cell layer.	[Ehrlich p. 182]
lymphoma	General term for malignancies affecting lymphoid tissues, including lymph nodes, spleen, liver, and bone marrow.	[Ehrlich p. 183]
Hodgkin’s lymphoma	Immune-system cancer distinguished by large cancerous lymphocytes called Reed-Sternberg cells.	[Ehrlich p. 183]

Term/Part	Meaning (textbook wording, concise)	Page
non-Hodgkin's lymphoma	All lymphomas other than Hodgkin's lymphoma; originate in lymphocytes and may be aggressive or indolent.	[Ehrlich p. 183]
breast cancer	Carcinoma developing from breast cells that can spread to adjacent lymph nodes and other sites.	[Ehrlich p. 183]
ER-positive	Describes breast cancers that grow in response to estrogen.	[Ehrlich p. 183]
BRCA1 and BRCA2 genes	Abnormal genes carrying higher risk of breast, ovarian, and certain other cancers.	[Ehrlich p. 183]
ductal carcinoma in situ	Breast cancer at its earliest stage, before breaking through the milk-duct wall.	[Ehrlich p. 183]
infiltrating ductal carcinoma	Starts in a milk duct and invades fatty breast tissue outside it; most common breast-cancer form.	[Ehrlich p. 183]
inflammatory breast cancer	Rare, aggressive breast cancer blocking lymphatic vessels in breast skin; rapid growth, pain, redness or rash, and nearby node swelling.	[Ehrlich p. 183]
male breast cancer	Occurs in breast tissue normally present in men; types resemble those in women.	[Ehrlich p. 183]

Diagnostic procedures

Term/Part	Meaning (textbook wording, concise)	Page
lymphoscintigraphy	Detects lymphatic-vessel damage or malformations by tracking an injected radioactive substance with a scanner or probe.	[Ehrlich p. 173]
bioimpedance spectroscopy	Noninvasive lymphedema diagnosis measuring resistance to electrical current through a limb; abnormally low results show lymph buildup.	[Ehrlich p. 173]
scratch test	Identifies troublesome allergens; swelling and itching indicate an allergic reaction.	[Ehrlich p. 174]
allergen-specific immunoglobulin E (IgE) blood test	Sometimes used to determine allergy to a particular substance.	[Ehrlich p. 174]
ELISA	Enzyme-linked immunosorbent assay; blood screening for HIV antibodies, Lyme disease, and other infectious conditions.	[Ehrlich p. 176]
western blot test	More accurate blood test used to confirm HIV diagnosis after a positive ELISA.	[Ehrlich p. 176]
staging	Classifying tumors by disease progression, potential response to therapy, and prognosis.	[Ehrlich p. 182]
breast self-examination	Self-care for early breast-cancer detection; checks for new lumps or changes in existing lumps, nipple shape, or skin.	[Ehrlich p. 183]; [Ehrlich p. 184]
palpation of the breast	Provider examination of breast texture, size, and consistency.	[Ehrlich p. 184]
mammography	Radiographic breast examination to detect tumors or precancerous cells.	[Ehrlich p. 184]
mammogram	Record resulting from mammography.	[Ehrlich p. 184]

Term/Part	Meaning (textbook wording, concise)	Page
molecular breast imaging	Nuclear medicine technique using gamma radiation to detect potential tumors in particularly dense breast tissue.	[Ehrlich p. 184]
ultrasound	Often an initial follow-up test for an abnormal mammography finding.	[Ehrlich p. 184]
breast MRI	Uses magnets, radio waves, and IV contrast for detailed information in diagnosed patients or cancer detection in high-risk patients.	[Ehrlich p. 184]
needle breast biopsy	X-ray- or MRI-guided needle removes small breast-tissue samples for diagnosis and treatment planning.	[Ehrlich p. 184]
surgical biopsy	Removal of a small tissue piece for examination to confirm a diagnosis.	[Ehrlich p. 184]
sentinel node biopsy	Biopsy of the first lymph node contacted by cancer cells leaving their organ of origin.	[Ehrlich p. 185]
lymph node dissection	Removal of all nodes in a major group to determine or slow cancer spread.	[Ehrlich p. 185]

Treatments

Term/Part	Meaning (textbook wording, concise)	Page
vaccine	Preparation containing an antigen made from whole or partial disease-causing organisms that have been killed or weakened.	[Ehrlich p. 172]
vaccination	Provides protection against disease; some conditions require periodic boosters to maintain effectiveness.	[Ehrlich p. 172]
compression and exercise	Common lymphedema treatment to control swelling and minimize infections; not a cure.	[Ehrlich p. 173]
antihistamines	Relieve or prevent allergy symptoms by preventing histamine's effects.	[Ehrlich p. 174]
antiretroviral drugs	Combination regimen for HIV; helps control the disease's impact on immunity but does not cure HIV.	[Ehrlich p. 176]
immunotherapy	Also biological therapy; disease treatment that stimulates or represses the immune response.	[Ehrlich p. 177]
allergy desensitization	Immunotherapy that represses sensitivity to a particular allergen.	[Ehrlich p. 177]
synthetic immunoglobulins	Also immune serum; post-exposure prevention against certain viruses, providing temporary immunity.	[Ehrlich p. 177]
synthetic interferon	Used to treat multiple sclerosis, hepatitis C, and some cancers.	[Ehrlich p. 177]
monoclonal antibodies	Laboratory-produced antibodies from identical offspring of a specific cell clone; enhance immune response to certain malignancies.	[Ehrlich p. 177]
immunosuppression	Treatment repressing or interfering with immune response to antigen stimulation.	[Ehrlich p. 177]
immunosuppressant	Substance preventing or reducing normal immune response; prevents donor-tissue rejection and depresses autoimmune disorders.	[Ehrlich p. 177]
corticosteroid drug	Hormone-like preparation used primarily as an anti-inflammatory and immunosuppressant.	[Ehrlich p. 177]

Term/Part	Meaning (textbook wording, concise)	Page
cytotoxic drug	Medication that kills or damages cells; used as an immunosuppressant or antineoplastic.	[Ehrlich p. 177]
measles, mumps, and rubella vaccination	MMR immunization can prevent these three viral illnesses; administered in early childhood.	[Ehrlich p. 179]
antibiotics	Medications inhibiting growth or killing pathogenic bacteria; not effective against viral infections.	[Ehrlich p. 181]
bactericide	Substance causing bacterial death; examples include penicillins and cephalosporins.	[Ehrlich p. 181]
bacteriostatic agent	Slows or stops bacterial growth; examples include tetracycline, sulfonamide, and erythromycin.	[Ehrlich p. 181]
antifungal	Also antimycotic; agent destroying or inhibiting fungal growth.	[Ehrlich p. 181]
antiviral drug	Used to treat viral infections or provide temporary immunity; acyclovir is an example.	[Ehrlich p. 181]
antiangiogenesis	Treatment disrupting a tumor’s blood supply.	[Ehrlich p. 181]
lumpectomy	Removal of cancerous tissue and a surrounding margin of normal tissue, leaving the remainder of the breast.	[Ehrlich p. 185]
mastectomy	Removal of the entire breast and nipple; often includes adjacent axillary nodes.	[Ehrlich p. 185]
radical mastectomy	Removal of an entire breast and many surrounding tissues.	[Ehrlich p. 185]
modified radical mastectomy	Removal of the entire breast and all axillary nodes under the adjacent arm.	[Ehrlich p. 185]
immediate breast reconstruction	Begins during mastectomy surgery with placement of an expander to replace removed tissue.	[Ehrlich p. 185]
delayed breast reconstruction	Restores the missing breast’s size and shape; may be necessary when radiation follows surgery.	[Ehrlich p. 185]
cancer surgery	Usually removes the malignancy and a margin of normal tissue; may remove nearby nodes to detect spread.	[Ehrlich p. 186]
laser surgery	Targeted light beams destroy cancer cells.	[Ehrlich p. 186]
cryosurgery	Freezes and destroys cancer cells using a substance such as liquid nitrogen.	[Ehrlich p. 186]
chemotherapy	Chemical agents and drugs used in combinations to destroy malignant cells and tissues.	[Ehrlich p. 186]
chemoprevention	Natural or synthetic substances reduce cancer development or recurrence risk, or reduce tumor size or slow development.	[Ehrlich p. 186]
antineoplastic	Medication blocking malignant-cell development, growth, or proliferation.	[Ehrlich p. 186]
radiation therapy	Used against some cancers with the goal of destroying cancerous tissue while sparing healthy tissue.	[Ehrlich p. 186]
brachytherapy	Radioactive materials placed in contact with or implanted into tissues being treated.	[Ehrlich p. 186]
teletherapy	Radiation therapy administered at a distance from the body.	[Ehrlich p. 186]

Term/Part	Meaning (textbook wording, concise)	Page
tomotherapy	Combines tomography with radiation therapy to target a tumor precisely, delivering radiation slice by slice.	[Ehrlich p. 187]
targeted therapy	Drugs or other substances identify and attack specific cancer cells without harming normal cells.	[Ehrlich p. 187]
adjuvant therapy	Treatment after primary cancer treatment to decrease recurrence risk.	[Ehrlich p. 187]
clinical trials	Testing promising cancer treatments not yet FDA-approved in patients agreeing to participate in research.	[Ehrlich p. 187]

Textbook vs course handout

- **Immun / immun/o:** The handout gives “Immun” as “immune.” The vocabulary list spells the combining form **immun/o** and includes “immune, protection, safe.” [Ehrlich p. 164]
- **Myel / myel/o:** The handout gives only “spinal cord.” The textbook gives **myel/o** for bone marrow and explicitly states that it also means spinal cord. [Ehrlich p. 163]
- **Leuk / leuk/o:** The handout writes “Leuk”; the textbook writes **leuk/o**. Both mean “white.” [Ehrlich p. 182]
- **My / my/o:** The handout writes “My.” The textbook uses **my** in myoma and **my/o** in myosarcoma; both mean “muscle.” [Ehrlich p. 181]
- **-cyte:** The handout gives “cells in.” The textbook defines **-cyte** as “cell” and **-cytes** as “cells”; it does not include “in.” [Ehrlich p. 172]; [Ehrlich p. 166]
- **-oma:** The handout gives “tumor of.” The textbook vocabulary list gives “tumor, neoplasm.” [Ehrlich p. 164]
- **Macro- / macro-:** The handout gives “large, abnormal size.” The textbook’s macrophage breakdown gives **macro-** as “large,” without “abnormal size.” [Ehrlich p. 172]

1. Word Parts

Blood & Immune Roots / Combining Forms

Part	Meaning	Example	Source
hem/o	blood	hemolysis	[general] ✓ **Ehrlich**
hemat/o	blood	hematocrit	[general] ✓ **Ehrlich**
erythr/o	red	erythrocyte	[handout] (listed as "Eryhr" — *typo; correct form is erythr/o*)
leuk/o	white	leukocyte	[handout]
thromb/o	clot; platelet	thrombocyte	[general] ✓ **Ehrlich**
cyt/o	cell	cytology	[general] ✓ **Ehrlich**
phleb/o	vein	phlebitis	[general] ✓ **Ehrlich**
ven/o	vein	venipuncture	[general] ✓ **Ehrlich**
lymph/o	lymph	lymphocyte	[general] ✓ **Ehrlich**
lymphaden/o	lymph node	lymphadenopathy	[general] ✓ **Ehrlich**
splen/o	spleen	splenomegaly	[general] ✓ **Ehrlich**
thym/o	thymus gland	thymectomy	[general] ✓ **Ehrlich**
tonsill/o	tonsils	tonsillitis	[general] ✓ **Ehrlich**
immun/o	immune; immunity	immunodeficiency	[handout]
path/o	disease	pathology	[general]
phag/o	to eat, devour	phagocyte	[general] ✓ **Ehrlich**
sider/o	iron	sideropenia	[general]

myel/o	bone marrow; also spinal cord	myelopoiesis	[handout] (handout lists "spinal cord"; in blood context = bone marrow)
plasm/o	plasma; formed substance	plasmapheresis	[general]
kary/o	nucleus	karyolysis	[general]
granul/o	granule	granulocyte	[general]
neutr/o	neutral (staining)	neutrophil	[general]
eosin/o	eosin (red dye); rose-colored	eosinophil	[general]
bas/o	base (basic-staining)	basophil	[general]
morph/o	shape; form	morphology	[general]
poikil/o	varied; irregular	poikilocytosis	[general]

Prefixes

Part	Meaning	Example	Source
aniso-	unequal; uneven	anisocytosis	[general]
an-	without; not	anemia	[handout]
pan-	all	pancytopenia	[general] ✓ Ehrlich
poly-	many; much	polycythemia	[general]
auto-	self	autoimmune	[general]
anti-	against	antibody	[general] ✓ Ehrlich
hyper-	excessive; above normal	hyperbilirubinemia	[handout]
hypo-	deficient; decreased	hypovolemia	[handout]
macro-	large; abnormal size	macrophage	[handout]
micro-	small	microcytic	[handout]

Suffixes

Part	Meaning	Example	Source
-cyte	cell	erythrocyte	[handout]
-emia	blood condition	anemia	[general] ✓ Ehrlich
-penia	deficiency; decrease	leukopenia	[general] ✓ Ehrlich
-poiesis	formation; production	hematopoiesis	[general]
-lysis	destruction; breakdown	hemolysis	[general]
-stasis	stopping; standing still	hemostasis	[general]
-osis	abnormal condition; disease	leukocytosis	[handout]
-philia	attraction to; affinity for	eosinophilia	[general]
-megaly	enlargement of	splenomegaly	[handout]
-pathy	disease	lymphadenopathy	[general] ✓ Ehrlich
-itis	inflammation of	phlebitis	[handout]
-oma	tumor of	lymphoma	[handout]
-rrhage	bursting forth; bleeding	hemorrhage	[handout]
-ectomy	surgical removal	splenectomy	[handout]
-tomy	cutting; incision	phlebotomy	[general]
-globin	protein (blood)	hemoglobin	[general] ✓ Ehrlich

-phage	one that eats	macrophage	[general] ✓ Ehrlich
-cytosis	increase in cells	leukocytosis	[general]
-apheresis	removal; to take away	plasmapheresis	[general]
-phylaxis	protection	prophylaxis	[general]

*Note: Handout lists "Demat" for skin — this is a typo; correct form is **derm/o** (from derm/at). Not blood/immune but flagged per course note.*

2. Blood Terms

Formed Elements & Key Components

Term	Meaning
erythrocyte (RBC)	Red blood cell; carries O_2 via hemoglobin/iron. Bone-marrow product.
leukocyte (WBC)	White blood cell; defends against infection/foreign material. Bone-marrow product.
thrombocyte (PLT)	Platelet; primary hemostasis (platelet plug). Bone-marrow product.
hemoglobin	Iron-containing protein inside RBCs that binds and carries O_2.
hematocrit	Percentage of total blood volume occupied by RBCs.
fibrinogen	Plasma protein; converted to fibrin during coagulation; also an inflammatory marker (helps locate pathogens).
bilirubin	Breakdown product of hemoglobin; transported to liver by albumin; excreted in bile; gives stool its brownish-yellow color.

Disorders & Conditions

Term	Meaning
anemia	Lower than normal RBCs or hemoglobin → tissues get less O_2. Symptoms: fatigue, weakness, pallor, dyspnea, tachycardia, dizziness, chest pain.
microcytic hypochromic anemia	Small (microcytic), pale (hypochromic) RBCs — iron deficiency. Low MCV + low MCHC on CBC.
macrocytic / megaloblastic anemia	Large, immature RBCs — B_{12} (cobalamin) or B_9 (folic acid) deficiency. Needed for DNA synthesis in rapidly dividing marrow.
pernicious anemia	B_{12} deficiency caused specifically by lack of intrinsic factor (IF) → B_{12} cannot be absorbed. *Only* use this name when IF is the problem.
aplastic anemia	Bone marrow failure; marrow not producing cell lines. Causes: radiation, toxic drugs/chemicals, marrow cancer.
hemolytic anemia	Excessive or premature RBC destruction (hemolysis). Causes: sickle cell, toxins, burns, mechanical valves, autoimmune attack. Breakdown of Hb → bilirubin → jaundice.
blood-loss anemia	Acute: normocytic/normochromic (whole blood lost). Chronic GI bleed: can become microcytic/hypochromic once iron stores are spent.
polycythemia	Too many RBCs. Causes: marrow disorder (polycythemia vera = blood cancer) or chronic hypoxia (smoking, COPD, high altitude). Effects: ↑ viscosity, ↑ volume.
leukocytosis	Increase in WBCs. Can be normal (post-op, MI) — not always infection. Monitor trends.

leukopenia	Decrease in WBCs.
neutropenia	Decrease in neutrophils (a type of leukopenia).
thrombocytopenia	Decrease in platelets.
thrombocytosis	Increase in platelets.
hemolysis	Destruction/bursting of RBCs (e.g., wrong-type transfusion: antigen meets its antibody → hemolysis).
hemostasis	Process of stopping bleeding. 5 stages: (1) vessel spasm/vasoconstriction, (2) platelet plug (primary hemostasis), (3) coagulation cascade (fibrinogen→fibrin; secondary hemostasis), (4) clot retraction, (5) fibrinolysis.
coagulation	Clot formation; cascade of clotting factors converting fibrinogen to fibrin.
embolus	Circulating clot/foreign material that lodges in and obstructs a vessel.
thrombus	Clot that forms within a vessel in situ (in place).
hemophilia	Inherited bleeding disorder; deficiency of a clotting factor → impaired coagulation.
hemochromatosis	Iron overload disorder (hemo- + chroma- + -osis); excess iron deposits in organs.
leukemia	Cancer of white blood cells; uncontrolled proliferation of abnormal WBCs in marrow/blood.
hypovolemia	Low blood volume (hypo = low, vol = volume, -emia = blood). Leads to ↓ BP → shock if severe. Observe: tachycardia *before* hypotension.
phlebotomy	Incision or puncture into a vein (blood draw).

Lab & Diagnostic Terms

Term	Meaning
CBC (complete blood count)	Reports hemoglobin, RBC count, platelets, WBCs, plus indices (MCV, MCHC).
MCV (mean corpuscular volume)	Size of the RBC. Low in microcytic anemia.
MCHC (mean corpuscular hemoglobin concentration)	How much Hb is packed in each cell. Low in hypochromic anemia.
differential (WBC differential)	Breakdown of WBC types: neutrophils, lymphocytes, monocytes, eosinophils, basophils.
ESR (erythrocyte sedimentation rate)	Inflammatory markers stick to RBCs → heavier → sediment faster. ↑ ESR = inflammation. *One of 3 markers your Patho instructor expects you to know.*
CRP (C-reactive protein)	Liver-made protein; inflammatory marker; regulates cytokines. *One of 3 markers expected.*
left shift	**Increase in immature neutrophils (bands).** Indicates marrow is releasing cells early. **NOT the same as neutrophilia** (which is simply an increase in total neutrophils).
reticulocyte count	Index of marrow activity. Reticulocytes = slightly immature RBCs (~1% of circulating). Low = lazy/blocked marrow or missing ingredients. High = marrow working hard (e.g., after hemorrhage).

3. Immune & Lymphatic Terms

Lymphatic Structures & Disorders

Term	Meaning
lymphadenopathy	Disease of lymph nodes (enlarged, abnormal nodes) — lymphaden/o + -pathy.
lymphadenitis	Inflammation of lymph nodes — lymphaden/o + -itis.
splenomegaly	Enlargement of the spleen — splen/o + -megaly.
splenectomy	Surgical removal of the spleen — splen/o + -ectomy.
lymphedema	Swelling (edema) caused by impaired lymph drainage.
lymphoma	Tumor (cancer) of lymphatic tissue — lymph/o + -oma.

Primary lymph organs: bone marrow & thymus (where immune cells are produced/matured). Secondary: spleen, lymph nodes, tonsils, appendix, Peyer's patches (GI), MALT (mucosa-associated lymphoid tissue). Losing a secondary organ (e.g., tonsils) is not the same as losing bone marrow.

Antigen vs Antibody

Antigen = a marker (surface molecule) on a cell membrane used to determine self vs non-self. **Not automatically bad** — self-antigens are left alone; foreign antigens are targeted. Also called antigenic determinant sites / epitopes.

Antibody = a protein (immunoglobulin) that the immune system *makes* to bind and neutralize a foreign antigen. Antigen + antibody = antigen-antibody (immune) complex.

Immunoglobulins (Antibody Classes)

Class	Key Facts
IgG	Most abundant. *Only Ig that crosses the placenta* — gives newborns protection for months. Long-lasting.
IgA	Secretory; in tears, saliva, GI & respiratory secretions. Protects mucous membranes (portals of entry).
IgM	First antibody to appear; suggests *current/recent* infection. Does not persist long.
IgE	Allergic reactions; defends against parasites. Mediates Type I hypersensitivity.
IgD	On B-cell membrane; stimulates B-cell proliferation when its antigen binds.

Immune Dysfunction

Term	Meaning
immunodeficiency	One or more branches of the immune system not working. Results: persistent/hard-to-treat infections, opportunistic infections, ↑ cancer incidence. Types: B-cell, T-cell, combined T+B, complement deficiency, phagocytosis disorders.
autoimmune / autoimmunity	Breakdown in self-tolerance; immune system attacks self-antigens → damage to self tissue. Often chronic with remissions and flares. Autoantibodies (e.g., ANA) are produced.

Hypersensitivity Types I–IV — Mnemonic ABCD

A = **A**llergies · **B** = anti**B**odies (cytotoxic) · **C** = immune **C**omplex · **D** = **D**elayed

Type	Mechanism	Timing	Examples
I — Allergies	IgE-mediated; mast cells/basophils degranulate → histamine, leukotrienes, bradykinin	~15 min	Allergic rhinitis (hay fever), asthma, urticaria (hives), atopic dermatitis (eczema), anaphylaxis

II — antiBodies (cytotoxic)	Antibody-mediated (IgG/IgM); antibody binds antigen on cell → binding itself damages/destroys cell	15–30 min	Blood transfusion reactions (e.g., type A recipient given type B blood)
III — immune Complex	Immune complex-mediated (IgG/IgM); complexes deposit in tissue → activate complement → inflammation	~6 hours	SLE (ANA binds DNA → complexes deposit in collagen-rich tissue)
IV — Delayed	T cell-mediated (no antibody); T cells sensitized → attack on re-exposure	1–3 days	Contact dermatitis (poison ivy), type 1 diabetes, certain transplant rejections

Key Immune / Allergy / Inflammation Terms

Term	Meaning
anaphylaxis	Most severe Type I reaction. Systemic degranulation → whole-body vasodilation → hypotension → can progress to anaphylactic shock. Also bronchoconstriction. Emergency: IM epinephrine first.
urticaria	Hives; raised, itchy skin wheals (like mosquito bites). Type I, IgE-mediated.
angioedema	Swelling of tongue, lips, part of face. Can accompany anaphylaxis or occur alone.
SLE (systemic lupus erythematosus)	Autoimmune; antigen-antibody complexes deposit in collagen-rich tissues → multi-system. Classic: butterfly/malar rash (worsened by sun). Dx = diagnosis of exclusion ("the great imitator").
ANA (antinuclear antibody)	Autoantibody directed against nuclear/DNA antigens. Primary autoantibody in SLE.
MHC (major histocompatibility complex)	Cell surface molecules (Class I & Class II) used for self-recognition. Part of self-tolerance maintenance.
degranulation	Release of inflammatory mediators (histamine, leukotrienes, bradykinin) from mast cell/basophil granules. = inflammation.
mast cells	Tissue cells similar to basophils; granules hold mediators of inflammation.
exudate	Fluid leaking from capillaries during inflammation. Types: serous (clear), serosanguineous (pink), purulent (pus). Usually beneficial — brings nutrients, O_2, antibodies.
acute-phase response	Systemic effects of inflammation: fever, leukocytosis, ↑ inflammatory markers (CRP, fibrinogen, ESR), skeletal muscle catabolism.
self-tolerance	Absence of immune response against one's own antigens. Maintained by: (1) regulatory T cells, (2) bone marrow/thymus screening of immature WBCs, (3) MHC molecules.

HIV/AIDS & Related

Term	Meaning
HIV (human immunodeficiency virus)	Retrovirus that destroys $CD4^+$ helper T cells → progressive loss of adaptive immunity.
AIDS (acquired immunodeficiency syndrome)	Advanced stage of HIV infection; defined by CD4 count < 200 cells/μL or presence of opportunistic infections / defining cancers.
CD4	Cluster of differentiation 4; membrane marker on helper T cells ("master switch"). Primary target of HIV.

opportunistic infection	Infection caused by organisms that typically do *not* cause disease in immunocompetent people; occurs when immune defenses are weakened (e.g., AIDS).
Kaposi sarcoma	Vascular tumor (associated with HHV-8); defining AIDS cancer. Presents as purple/red skin lesions.

Phagocytes & Innate Immune Cells

Term	Meaning
phagocyte	Cell that engulfs and digests foreign particles, tumor cells, virus-infected cells. Includes neutrophils and monocyte-macrophages.
macrophage	Large phagocytic cell (monocyte-macrophage lineage). Engulfs debris and acts as antigen-presenting cell (APC).
natural killer (NK) cells	Innate lymphocyte; kills tumor and virus-infected cells by releasing toxins (not phagocytosis). Does *not* need helper T activation.
cytokines	Hormone-like proteins released by WBCs to regulate immune response. Examples: interferons, interleukins, TNF, colony-stimulating factor. Can produce fever (act on hypothalamic thermoregulatory center).
complement system	Cascade of ~20 plasma proteins. Enhances phagocytosis/antibodies. Key results: degranulation, chemotaxis, lysis, opsonization. Key word to remember: **inflammation**.

4. Patho Vocabulary Tie-Ins

Terms from Pathophysiology lectures — Blood (Hematopoietic), Immune 1, Immune 2. One-line meaning; class noted. Follow these definitions for the quiz.

Term	Meaning	Class
hematopoiesis	Formation of blood cells from hematopoietic stem cells in red bone marrow.	Blood
erythropoiesis	Formation of RBCs. Stimulus: hypoxia → kidney makes EPO → marrow makes RBCs.	Blood
erythropoietin (EPO)	Kidney hormone; stimulates RBC production in response to hypoxia.	Blood
reticulocyte	Slightly immature RBC released from marrow (~1% of circulating RBCs). Retic count = index of marrow activity.	Blood
normocytic normochromic	Normal size, normal color RBCs — seen in acute blood loss (whole blood lost).	Blood
microcytic hypochromic	Small, pale RBCs — iron deficiency anemia. Low MCV + low MCHC.	Blood
macrocytic / megaloblastic	Large, immature RBCs — B_{12} or folate deficiency.	Blood
intrinsic factor (IF)	Made by gastric parietal cells; required for B_{12} absorption. Lack of IF → pernicious anemia.	Blood
pernicious anemia	B_{12} deficiency *specifically* due to lack of IF. Not all B_{12} deficiency = pernicious.	Blood
polycythemia vera	Blood cancer; marrow overproduces RBCs. One cause of polycythemia.	Blood
hypovolemia	Low blood volume. Observe: tachycardia first, then hypotension.	Blood
MCV	Mean corpuscular volume — size of RBC.	Blood

MCHC	Mean corpuscular hemoglobin concentration — Hb per cell.	Blood
bilirubin	Hb breakdown product; transported to liver by albumin; excreted in bile.	Blood
cheilosis / cheilitis	Cracked, inflamed corners of the mouth — clue for B-vitamin or iron deficiency.	Blood
paresthesias	Numbness, tingling — B_{12} deficiency neuro change.	Blood
ataxia	Coordination problems — B_{12} deficiency neuro change.	Blood
left shift	**Increase in immature neutrophils/bands.** NOT the same as neutrophilia.	Blood
innate immunity	Natural/nonspecific; present at birth. Barriers, phagocytes, NK cells, cytokines, complement.	Immune 1
adaptive immunity	Specific/acquired; learned. T cells (cell-mediated) & B cells (humoral/antibody).	Immune 1
antigen	Marker on cell membrane for self vs non-self. Not inherently bad.	Immune 1
antibody	Protein made by immune system against a foreign antigen.	Immune 1
phagocytosis	Engulfing and digesting foreign particles/dead cells.	Immune 1
APC (antigen-presenting cell)	Processes foreign antigen and displays it for T cells.	Immune 1
helper T (CD4)	"Master switch" — activates other T cells and B cells via cytokines.	Immune 1
cytotoxic T (CD8)	Kills infected/tumor cells by releasing toxins. Must be activated by helper T first.	Immune 1
regulatory/suppressor T	Shuts off immune response; suppresses self-reactive WBCs.	Immune 1
opsonization	Coating antigen-antibody complexes with complement → easier phagocytosis.	Immune 1
active immunity	Your immune system generates the response (vaccine = antigen → you make antibody). Slow onset, long-lasting.	Immune 1
passive immunity	Received antibodies/immune cells from another source (gamma globulin = antibody). Immediate, short-lived.	Immune 1
self-tolerance	No immune response against own antigens. Maintained by: regulatory T cells, bone marrow/thymus screening, MHC molecules.	Immune 2
autoantibody	Antibody directed against self-antigens (e.g., ANA in SLE).	Immune 2
hypersensitivity	Excessive immune response to an antigen most people would not react to → unnecessary inflammation/damage.	Immune 2
sensitization	Process of becoming reactive to an allergen after first exposure.	Immune 2
plasmapheresis	Removal of plasma; used as intervention for autoimmune disease (removes autoantibodies).	Immune 2
CRP	C-reactive protein; liver-made inflammatory marker.	Immune 2
ESR	Erythrocyte sedimentation rate; markers stick to RBCs → sediment faster. Inflammation marker.	Immune 2
skeletal muscle catabolism	Breakdown of muscle protein → amino acids for antibody production/tissue repair.	Immune 2

leukocytosis	Increase in WBCs — part of acute-phase response. Not always infection (post-op, MI).	Immune 2

5. Word-Building Drills (25 Terms)

Cover the right two columns and break each one down yourself.

#	Term	Breakdown	Meaning
1	**hemolysis**	hem/o (blood) + -lysis (destruction)	destruction of red blood cells
2	**erythrocyte**	erythr/o (red) + -cyte (cell)	red blood cell
3	**leukocytosis**	leuk/o (white) + cyt (cell) + -osis (abnormal condition)	abnormal increase in WBCs
4	**leukopenia**	leuk/o (white) + -penia (deficiency)	decrease in WBCs
5	**thrombocytopenia**	thromb/o (clot) + cyt/o (cell) + -penia	decrease in platelets
6	**pancytopenia**	pan- (all) + cyt/o (cell) + -penia	decrease in all blood cell types
7	**anemia**	an- (without) + -emia (blood condition)	lack of RBCs/hemoglobin
8	**polycythemia**	poly- (many) + cyt (cell) + hem (blood) + -ia	too many RBCs
9	**hemostasis**	hem/o (blood) + -stasis (stopping)	stopping of bleeding
10	**hematopoiesis**	hemat/o (blood) + -poiesis (formation)	blood cell formation
11	**erythropoiesis**	erythr/o (red) + -poiesis	RBC formation
12	**phlebotomy**	phleb/o (vein) + -tomy (incision)	incision/puncture of a vein to draw blood
13	**hemorrhage**	hem/o (blood) + -rrhage (bursting forth)	excessive bleeding
14	**neutropenia**	neutr/o (neutral) + -penia	decrease in neutrophils
15	**eosinophilia**	eosin/o (rose dye) + -philia (attraction)	increase in eosinophils
16	**anisocytosis**	aniso- (unequal) + cyt (cell) + -osis	RBCs of unequal size
17	**poikilocytosis**	poikil/o (irregular) + cyt (cell) + -osis	RBCs of irregular shape
18	**lymphadenopathy**	lymph/o + aden/o (gland) + -pathy (disease)	disease/enlargement of lymph nodes
19	**lymphadenitis**	lymphaden/o (lymph node) + -itis	inflammation of lymph nodes
20	**splenomegaly**	splen/o (spleen) + -megaly (enlargement)	enlarged spleen
21	**splenectomy**	splen/o + -ectomy (surgical removal)	removal of the spleen
22	**thymectomy**	thym/o (thymus) + -ectomy	removal of the thymus
23	**lymphoma**	lymph/o + -oma (tumor)	tumor of lymphatic tissue

24	**immunodeficiency**	immun/o (immune) + deficiency	immune system not working fully
25	**phagocyte**	phag/o (eat) + -cyte (cell)	cell that engulfs and digests

6. Common Test Traps

- Spell **erythr/o** with both r's and the h: erythrocyte. The handout's "Eryhr" is a typo. Spelling counts on this quiz.
- **-penia** = too few; **-cytosis** / **-philia** = too many. Leukopenia ≠ leukocytosis.
- **-emia** = blood condition (anemia, polycythemia, leukemia). Leukemia is a cancer, not just high WBCs.
- **-lysis** = breakdown (hemolysis); **-stasis** = stopping (hemostasis). Easy to swap.
- **-poiesis** spelling: p-o-i-e-s-i-s (hematopoiesis, erythropoiesis).
- **-tomy** = cutting into (phlebotomy); **-ectomy** = removal (splenectomy); **-ostomy** = making an opening.
- **lymphaden/o** = lymph *node*; **lymph/o** = lymph. Lymphadenopathy is about nodes.
- **Antigen** = the marker on the cell; **antibody** = the protein made against it. Her #1 mix-up.
- **thromb/o** means clot; thrombocyte = platelet. **Thrombus** stays put; **embolus** travels.
- **myel/o** = bone marrow in blood terms (spinal cord in nervous-system terms).
- **Left shift** = more immature neutrophils (bands). It is *not* neutrophilia.
- Hypersensitivity order is **ABCD**: I Allergies (IgE), II antiBodies, III immune Complex, IV Delayed (T cell).
- **Pernicious** anemia only when intrinsic factor is missing. Not every B_{12} deficiency counts.
- **Auto-** = self (autoimmune); **anti-** = against (antibody); **ante-** = before. Don't mix them up.
- **hem/o** and **hemat/o** both mean blood: hematocrit, hematoma, hemoglobin.